

Report Homework #3

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Star Schema and Interactive Dashboard for Financial Transactions

1. Modeling choices

The business process is the trading activity of a brokerage account during 2024. The assignment already fixes the grain: one row of Fact_Transactions is one transaction of the account statement, so the fact table is an event fact with the same granularity of the source. Around the fact there are four dimensions (Dim_Time, Dim_Geography, Dim_Symbol and Dim_TransactionType) and one measure, Unit, which is a flow measure and can be summed along all the dimensions. Many questions ask for the number of transactions: for those I simply count the fact rows, so a second measure was not needed.

All the dimension tables have a surrogate key. I also added a surrogate key to the fact table (Transaction_SK), because at this grain two different transactions can have the same date, symbol and type, so the combination of the foreign keys is not unique. I could not use IDTransaction as key either, since in the source file it has a lot of duplicates; I kept it only as a descriptive attribute.

For the time hierarchy I used Day, Month, Quarter, Year, plus DayOfWeek as a separate branch of the Day level (one of the questions asks about Mondays). For the securities the hierarchy is Symbol, Industry, Sector, after checking that each industry belongs to only one sector. For geography the assignment says Country, Region, Sub-region, but in country.csv it is the opposite: every sub-region belongs to one region (Southern Europe is part of Europe, for example), so sub-region is the finer level and I used Country, Sub-region, Region. The company country is stored only in Dim_Geography, to avoid having the same attribute in two dimensions; the fact reaches the geography through the symbol.

2. Data quality issues found

- 464 completely empty rows in the account statement, plus an empty column (Unnamed: 5) created by the final ";" of every line. Both removed.
- The account statement and symbols.csv start with an invisible character that ends up attached to the name of the first column. Reading the files with encoding="utf-8-sig" removes it.
- The type DIVIDENT is clearly a typo for DIVIDEND, so I fixed the label. The dividend rows stay in the fact table, because the grain says one row per transaction.
- IDTransaction is not unique: 1,145 duplicated values on transactions that are clearly different. I did not delete anything, but the attribute cannot work as a key.
- 18 traded symbols (212 transactions, around 9%) are missing from symbols.csv. Without the master data I cannot fill the symbol and geography foreign keys, so I excluded these rows and documented the number. 2,069 valid transactions remain.
- Taiwan and Turkey are written differently in country.csv (Taiwan, Province of China and Türkiye), so I translated the two values during the integration.
- In country.csv the Taiwan row has region and sub-region empty. I filled them with Asia / Eastern Asia, otherwise the Taiwanese transactions would disappear from every analysis by region.

3. Main analytical results

Here are the main things that came out of the eight questions:

- The portfolio is clearly unbalanced towards US technology stocks: Technology is the first sector by SELL transactions in the US (158, almost three times the second one), and it stays first even looking only at Mondays.
- The activity is concentrated at the beginning of the year: Q1 alone has 968 of the 1,973 BUY + SELL operations, and then the volume drops quarter after quarter (Q3 and Q4 around 240 each).
- By country the United States dominates with 389 SELL transactions, before United Kingdom and China; at region level only Americas, Europe and Asia appear, and the Americas take more than half of the units bought.
- In Asia the first industry by traded units is Semiconductors (5,401), pushed by the Taiwanese stocks — without the fix on the Taiwan region this result would have been lost.
- The ranking by average units per transaction is led by Paysafe (248 units per operation in Q2), a name that almost never shows up in the count-based rankings: averages and counts answer different questions.

As a last check, the 1,973 BUY + SELL operations plus the 96 dividends give exactly the 2,069 rows of the fact table, so loading and queries are consistent.